User Layer
CLIENT
Home User
End-user accessing the AIRA platform via web or mobile browsers.
Web Browser
Mobile Browser
Presentation Layer
FRONTEND
React Dashboard
Responsive dashboard for energy monitoring, AI chat, and appliance
React
Tailwind CSS
Recharts
Firebase Auth
Backend & Logic Layer
API GATEWAY
API Gateway
Entry point for all frontend requests, handling routing and rate limiting.
Azure API Management
Nginx
SERVICE
AIRA Backend Service
Core logic for user sessions, energy data processing, and service
Node.js
Express
Firebase Admin SDK
Data Persistence Layer
DATABASE
AIRA Primary Database
Scalable NoSQL storage for user profiles, appliance data, and
MongoDB Atlas
Firebase Firestore
AI & Analytics Layer (Intelligence)
SERVICE
AI Recommendation Eng...
Advanced AI engine for pattern detection, consumption prediction,
Python
LangGraph
Azure OpenAI
Scikit-learn
Pandas
SERVICE
Energy Analytics Service
Processes raw energy data into visual insights, carbon footprint
Node.js
D3.js
Apache Spark
IoT & Edge Layer (Future Enhancement)
CLIENT
Smart Home IoT Devices
Hardware layer including ESP32, smart plugs, and solar sensors for
ESP32
Arduino
Zigbee
Wi-Fi
SERVICE
IoT Hub Gateway
Managed IoT hub for secure communication with smart meters
Azure IoT Hub
MQTT